CrimeLens: A Unified Residual Spatio-Temporal Graph Attention Network and Computer Vision Framework for Proactive Urban Safety

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***Abstract*—CrimeLens is a next-generation, web-based intelligence platform designed to revolutionize urban safety through the integration of spatio-temporal graph neural networks, real-time computer vision, and geospatial artificial intelligence. Designed for law enforcement, researchers, and community stakeholders, the platform automates the transformation of complex, high-volume municipal datasets into multidimensional, actionable insights. The flagship innovation of this system is the implementation of a Residual Spatio-Temporal Graph Attention Network (RST-GAT). Unlike traditional tabular machine learning models, this advanced predictive engine maps the physical layout of the city as an interconnected topological graph, processing six-month historical windows to forecast localized crime surges across precise urban grid cells. To bridge the gap between deep learning research and practical application, the system algorithmically translates these complex tensor outputs into a normalized, highly actionable 0-100 Risk Index for priority zones. This predictive core is complemented by immersive 3D Geospatial Density Mapping using WebGL. Beyond historical analysis, CrimeLens features an active Real-Time Surveillance Module utilizing YOLOv11 for weapon detection and a custom MobileNetV2 + LSTM architecture for complex violence recognition in live video feeds. Furthermore, an NLP-driven Public Perception Engine analyzes local news sentiment to calculate a community "Fear Index." Built on a decoupled, high-performance architecture using a FastAPI backend and a React.js frontend, the platform ensures rapid scalability and near-zero latency. Performance evaluations demonstrate a Mean Absolute Error (MAE) of 0.3924 for the RST-GAT spatial forecast, alongside a 97.1% weapon detection precision, marking a significant transition from reactive incident reporting to proactive, data-driven defense.**

***Index Terms*—Residual Spatio-Temporal Graph Attention Network (RST-GAT), Graph Neural Networks (GNN), Crime Risk Prediction, 3D Geospatial Visualization, YOLOv11, FastAPI, React.js.**

I. Introduction

Crime has remained a persistent challenge throughout human history, evolving in tandem with the structure of our societies. From early civilizations where theft and violence were direct consequences of resource scarcity, to the modern era of organized syndicates and cybercrime, criminal activity consistently reflects broader social and economic shifts. The Industrial Revolution marked a critical turning point; rapid urbanization and mass migration led to overcrowded cities, creating stark socioeconomic disparities that fostered property crimes and social unrest.

By the mid-2020s, the complexity of urban crime reached a new peak. While global statistics suggest a general decline in some non-violent categories, localized violent crime within densely populated urban centers has seen a concentrated surge. In India, data provided by the National Crime Records Bureau (NCRB) reflects a similar struggle. Major metropolitan hubs grapple with a high volume of assaults and sophisticated thefts driven by population density, wealth inequality, and rapid infrastructural expansion. Theoretical frameworks in criminology, such as the Routine Activity Theory, posit that crime occurs when three specific elements converge in time and space: a motivated offender, a suitable target, and the absence of a capable guardian. Conversely, the Crime Pattern Theory emphasizes how perpetrators tend to function inside well-known environmental geometries. CrimeLens is rooted in the belief that advanced deep learning can decipher these geometries, acting as a proactive "digital guardian."

A. Problem Statement and Challenges

Unpredictability and the sheer volume of incidents form the crux of the issue in urban environments. Traditional crime analysis systems rely heavily on reactive methodologies and static reporting. While existing systems, such as India's Crime and Criminal Tracking Network & Systems (CCTNS), are vital for archival data storage, they frequently lack the advanced visualization and predictive layers needed for proactive intervention.

Crucially, traditional predictive models suffer from severe architectural limitations. Time-series forecasting models (like ARIMA, SARIMA, or standard LSTMs) and flat-data classifiers (such as Random Forest or XGBoost) treat geographic zones as completely isolated data points. They fundamentally fail to capture the spatial "ripple effect"—the reality that a crime surge in one neighborhood heavily influences the risk profile of adjacent blocks. Furthermore, the disorganized state of raw crime data makes thorough analysis difficult. Manual reviews are time-consuming and prone to human partiality, exacerbated by frequent under-reporting in marginalized communities. The "functional split" between vast data repositories and actionable intelligence leaves law enforcement one step behind evolving crime patterns. To solve these fundamental limitations, CrimeLens shifts the paradigm from traditional machine learning to advanced spatio-temporal deep learning by introducing the RST-GAT as its core predictive engine.

B. Objectives and Scope

The primary objective of CrimeLens is to engineer a unified, AI-driven ecosystem that bridges the gap between historical data archives and proactive public safety. To achieve this, the project focuses on four critical pillars:

- **Advanced Spatio-Temporal Predictive Modeling:** Delivering state-of-the-art predictive foresight through the engineering of an RST-GAT. This involves algorithmically translating complex neural tensors into an intuitive 0-100 Risk Index to guide tactical police deployments.
- **Geospatial Intelligence:** Revolutionizing visualization by moving beyond static 2D maps to interactive Choropleth grids and fully immersive 3D Hexbin Density layers where elevation represents crime volume and color indicates severity.
- **Automated Surveillance:** Integrating high-speed computer vision pipelines (YOLOv11 and MobileNetV2+LSTM) for immediate, localized threat detection in public camera feeds.
- **Human-Centric AI & Sentiment Analysis:** Prioritizing decision support through a Gemini-powered AI Safety Assistant and an NLP-driven Public Perception Engine that scrapes news to gauge the psychological safety of a community.

II. Literature Survey

This chapter comprehensively reviews existing studies related to crime analytics, spatial forecasting, and predictive modeling, identifying the critical technological gaps that establish the theoretical foundation for the proposed RST-GAT architecture.

A. Traditional Machine Learning and Hotspot Detection

Detecting hotspots has historically been the fundamental method in crime analytics. Popular clustering algorithms include unsupervised methods like K-means and DBSCAN to assess high-density crime hotspots within a certain temporal and spatial boundary. A 2023 study by Bonam et al. [5] utilized optimized K-means clustering on municipal crime datasets, establishing the Elbow Method as a credible option for determining optimum spatial clusters, yielding 97% accuracy when used in tandem with Random Forests. Similarly, research by B. G and K. G [6] demonstrated the efficacy of flat machine learning classifiers in categorizing these pre-defined hotspots.

However, relying solely on basic geometric clustering often neglects critical environmental factors and urban topologies. Multi-density methods handling urban heterogeneity have emerged to address this. Cesario et al. [1] suggested combining multi-density clustering with regressive algorithms to forecast criminal activities in multi-density crime hotspots, proving that dynamic spatial boundaries yield more accurate results than rigid geometric grids.

B. Time-Series Forecasting and Statistical Regression

Forecasting methods have evolved toward cross-statistical and hybrid models. Studies utilizing exploratory data analysis (EDA) combined with XGBoost for predicting crime types and Holt-Winters Exponential Smoothing (HWES) demonstrated low root-mean-squared errors (RMSE) for macro-level 5-year trend forecasts [2]. Iqbal et al. [7] provided a comparative analysis of multiple regression methods for predicting specific types of crime in Chicago, noting that while traditional regressors perform well on aggregate city-wide data, their accuracy plummets when applied to granular, neighborhood-level predictions.

To combat this, short-term forecasting incorporating human mobility flows has shown significant F1-score improvements. Wu et al. [4] demonstrated that incorporating external features such as public transit flows and point-of-interest (POI) density into deep learning architectures substantially enhances short-term crime prediction. Furthermore, systematic reviews by Mandalapu et al. [9] and Maurya et al. [10] consistently highlight the transition from standard statistical models toward advanced neural architectures to capture non-linear temporal dependencies.

C. The Rise of Spatio-Temporal Graph Neural Networks

Recent advancements in urban computing have established Spatio-Temporal Graph Neural Networks (STGNNs) as the modern standard for predictive modeling. Unlike standard Convolutional Neural Networks (CNNs) that require data in a strict Euclidean grid (like pixels in an image), Graph Neural Networks (GNNs) can process data formulated as complex, irregular networks [8].

A comprehensive survey by Jin et al. [15] detailed how STGNNs excel in predictive learning for urban computing by treating city intersections as nodes and road networks as edges. More specifically applied to criminology, Wang et al. [16] introduced a framework utilizing STGNNs for accurate crime prediction, empirically proving that mapping urban environments as interconnected spatial adjacency graphs significantly outperforms conventional flat classifiers in both accuracy and recall by capturing the "contagion effect" of criminal behavior across adjacent districts. CrimeLens builds directly upon this paradigm by introducing residual connections to stabilize deep graph attention layers.

D. Computer Vision in Automated Surveillance

Automated security surveillance has transitioned from traditional computer vision to sophisticated deep learning frameworks to combat the global crisis of gun violence and public safety threats. Early methodologies relied on hand-crafted features (such as SIFT, SURF, and Harris) [13], which achieved accuracies between 84% and 88% but struggled significantly with the inconspicuous nature of small handguns and varying urban lighting conditions.

Modern research overwhelmingly favors the You Only Look Once (YOLO) series due to its ability to process images in a single pass, making it viable for real-time video. A 2024 study by Thakur et al. [12] implemented real-time weapon detection using YOLOv8, identifying high-speed tracking solutions for enhanced safety. CrimeLens extends this by utilizing the latest YOLOv11 framework coupled with ByteTrack for sustained object persistence.

For complex violence detection, static bounding boxes are insufficient. Spatio-temporal Networks (STNs) combine CNNs with Recurrent Neural Networks (RNNs) to capture both spatial features and kinetic sequences. Vosta and Yow [11], [14] demonstrated that a CNN-RNN combined structure utilizing feature extraction followed by Convolutional LSTM (ConvLSTM) preserves spatial relationships while identifying anomalies in time-series data, achieving state-of-the-art Area Under Curve (AUC) metrics on benchmark surveillance datasets.

E. Gaps in Existing Research

Despite the massive advancements across these domains, a critical review reveals several fundamental gaps:

- **The Fallacy of Spatial Independence:** The vast majority of deployed predictive policing systems still rely on tabular models (like Random Forests) that inherently treat geographic zones as isolated data points, ignoring the dynamic, physical flow of risk across city streets.
- **Integration Chasm:** Current academic literature treats predictive data analytics, computer vision surveillance, and NLP sentiment analysis as strictly separate, siloed research domains [3]. There is a distinct lack of platforms that unify these models into a single, cohesive operational pipeline.
- **The "Black Box" Problem:** Advanced deep learning models frequently output fractional tensors that are theoretically accurate but operationally useless for law enforcement personnel who require clear, explainable, and localized risk indices.

CrimeLens was designed specifically to bridge these gaps, uniting disparate AI disciplines under a highly scalable, web-based architecture.

III. System Architecture

The architecture of CrimeLens is built on a high-performance, decoupled framework designed to bridge the gap between large-scale historical data analysis and real-time security monitoring. As depicted in Figure 1, the system utilizes a multi-layered approach to ensure scalability, fault tolerance, and minimal latency during heavy tensor processing.

The architecture is broadly classified into four active, asynchronous zones:

1) **Client Presentation Layer (React.js):** A highly responsive Single Page Application (SPA) managing state across interactive 3D geospatial maps, Chat LLM interfaces, and active WebRTC video streaming buffers.
2) **API Gateway & Orchestration (FastAPI):** An asynchronous routing hub that securely controls incoming payloads, manages session states, and strictly enforces data validation using Pydantic models.
3) **Predictive Intelligence Engine (PyTorch/Scikit-Learn):** An isolated computational environment where heavy matrix multiplications occur for spatial grid generation, graph attention calculations, and historical regression modeling.
4) **Active Surveillance Module (Computer Vision):** A multi-threaded environment where localized camera feeds are buffered, segmented, and evaluated by the YOLO and LSTM models without blocking the API Gateway.

A. Microservice Decoupling

To guarantee that the user dashboard does not freeze while the backend processes massive mathematical graphs, CrimeLens relies entirely on FastAPI's `asyncio` event loop. Data intensive tasks (such as training the XGBoost classifier or aggregating hundreds of thousands of raw coordinates) are dispatched to background threads. This ensures that the primary Web Server Gateway Interface (ASGI) remains unblocked and highly responsive to incoming HTTP requests.

IV. Methodology and Mathematical Framework

The implementation of CrimeLens follows a rigorous, mathematically sound methodology integrating advanced data engineering, spatio-temporal deep learning, and multi-modal computer vision.

A. Data Ingestion, Cleaning, and Preprocessing

Real-world municipal crime datasets are inherently noisy, sparse, and inconsistent. The ingestion pipeline utilizes the `Pandas` library via the FastAPI backend to sanitize this data. The pipeline standardizes temporal strings by applying a Z-fill to incomplete four-digit times (e.g., coercing "830" into "0830") and parsing inconsistent localized date formats into unified ISO-8601 datetime objects.

Missing geographic data is imputed or dropped, and invalid zero-coordinate entries (frequently used by dispatchers as placeholders) are filtered out. Numerical features are normalized using Min-Max scaling to ensure stability during gradient descent:

$$x'_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \quad (1)$$

Temporal features such as the specific Hour, Month, and Day of the Week are engineered from the raw timestamps to provide cyclic context to the learning algorithms.

B. Spatial Structuring and Grid Formation

To prepare the scattered point data for processing by the Graph Neural Network, the system must first map the continuous urban landscape into a discrete mathematical structure.

Let the geographic bounding box of the city be defined by the coordinate limits $[Lat_{min}, Lat_{max}]$ and $[Lon_{min}, Lon_{max}]$. The region is divided into a uniform matrix of spatial grid cells with a designated resolution r (e.g., $r = 0.05$ degrees of latitude/longitude).

Each individual criminal incident $p_k = (lat_k, lon_k)$ is mathematically mapped to a unique grid cell identifier $C_{i,j}$

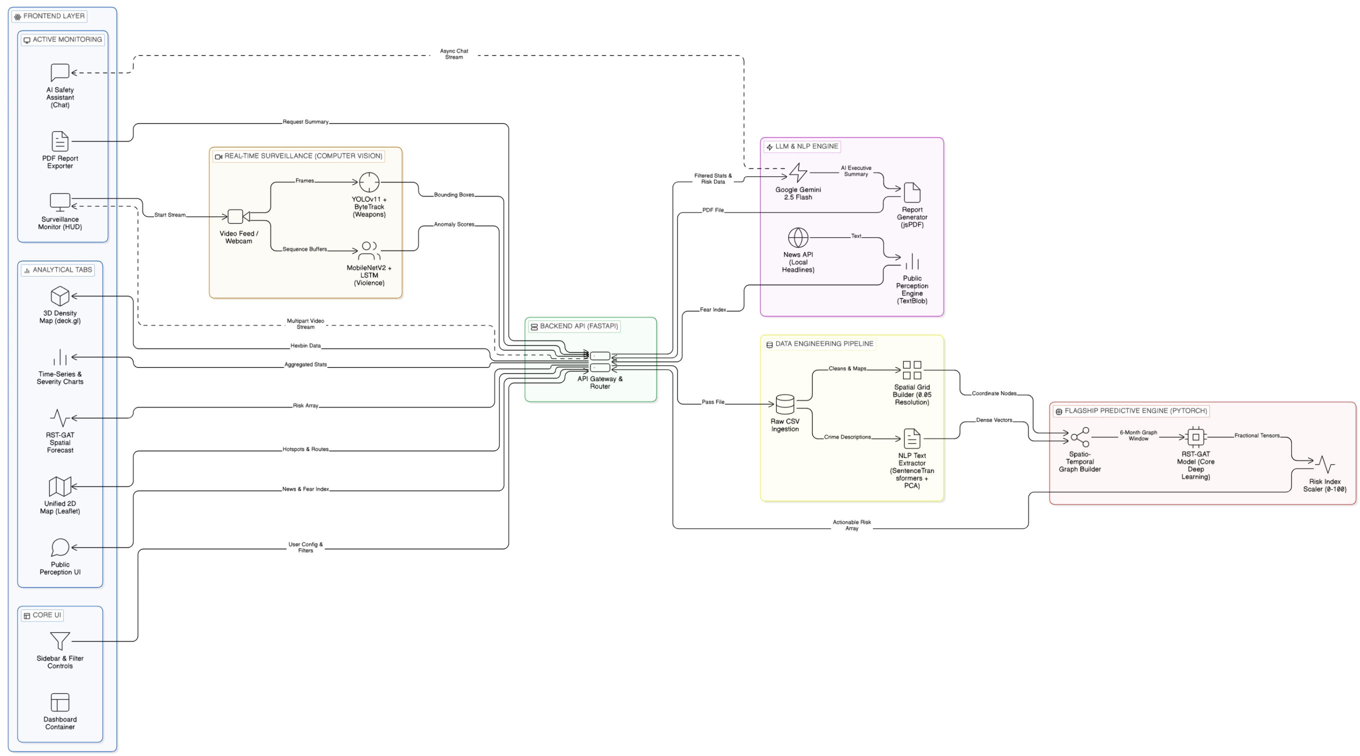

Fig. 1. System Architecture of the CrimeLens Platform. The design enforces strict separation of concerns, allowing asynchronous data ingestion and computationally heavy deep learning inference to occur without bottlenecking the primary user interface.

based on floor division of its coordinates. This aggregates raw coordinate points into actionable neighborhood zones, converting a massive list of incidents into a structured frequency distribution across the city.

C. NLP Feature Extraction via PCA

Categorical string data (e.g., "AGGRAVATED ASSAULT WITH DEADLY WEAPON") cannot be natively processed by the mathematical layers of the graph neural network. To capture the nuance of these crimes, the pipeline utilizes the `SentenceTransformers` library to encode textual crime descriptions into dense semantic numerical vectors of size 384.

Because storing a 384-dimensional vector for every single grid cell over multiple months would exceed GPU VRAM constraints, Principal Component Analysis (PCA) is applied to reduce the dimensionality to $D = 32$. This creates a highly optimized, dense feature tensor that accurately represents the semantic nature of crimes occurring within each spatial grid.

D. Graph Construction and Adjacency Modeling

The core theoretical advancement of CrimeLens relies on modeling the city as a dynamic, interconnected network rather than a series of isolated tables. The urban environment is formalized as an unweighted, undirected spatial graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

The nodes $\mathcal{V} = \{v_1, v_2, ..., v_N\}$ represent the N active grid cells in the city. The edges $\mathcal{E}$ represent the spatial adjacency between these grids. Two grid cells v_i and v_j are connected by an edge if the Haversine distance between their spatial centroids is below a predefined threshold ϵ. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is constructed as:

$$A_{i,j} = \begin{cases} 1 & \text{if } \text{dist}_{Haversine}(v_i, v_j) \leq \epsilon \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

This matrix $\mathbf{A}$ allows the neural network to mathematically interpret which neighborhoods physically border each other, enabling the flow of risk intelligence across boundaries during the convolution phase.

E. Flagship Predictive Engine: RST-GAT

Moving beyond traditional flat-data machine learning, the system deploys a Residual Spatio-Temporal Graph Attention Network (RST-GAT). The model takes the input feature tensor $\mathbf{X} \in \mathbb{R}^{N \times T \times F}$ (where N is nodes, T is historical time windows, and F is the feature dimension) and the adjacency matrix $\mathbf{A}$.

1) Multi-Head Spatial Attention Mechanism: To process this graph, the model applies a Graph Attention (GAT) layer. Unlike basic Graph Convolutional Networks (GCNs) which treat all neighboring nodes equally, the GAT dynamically learns to weigh the varying influence of neighboring grids. For instance, a grid containing a major highway may exert a higher risk influence on its neighbors than a grid containing a quiet park.

First, a shared linear transformation, parameterized by a weight matrix $\mathbf{W} \in \mathbb{R}^{F' \times F}$, is applied to every node. A self-attention mechanism a then computes the attention coefficients indicating the importance of node j's features to node i. To make coefficients easily comparable across different nodes, they are normalized using the softmax function. To prevent equation overflow within the formatted columns, this is expressed using a dynamically resized box:

$$\alpha_{i,j} = \frac{\exp\Big(\text{LeakyReLU}\Big(\vec{\mathbf{a}}^T [\mathbf{W}\vec{h}_i || \mathbf{W}\vec{h}_j]\Big)\Big)}{\sum_{k \in \mathcal{N}_i} \exp\Big(\text{LeakyReLU}\Big(\vec{\mathbf{a}}^T [\mathbf{W}\vec{h}_i || \mathbf{W}\vec{h}_k]\Big)\Big)} \quad (3)$$

The hidden state of node i at the next layer is then computed as a non-linear combination of its neighbors' weighted features:

$$\vec{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \alpha_{i,j} \mathbf{W} \vec{h}_j^{(l)} \right) \quad (4)$$

To stabilize the learning process, multi-head attention is employed, calculating K independent attention mechanisms and concatenating their features.

2) Residual Temporal Fusion: A common flaw in deep graph architectures processing long temporal windows is the "over-smoothing" phenomenon, where repeated neighborhood aggregations cause all node embeddings to converge to the same value, erasing localized risk disparities.

To counteract this, CrimeLens introduces a novel residual temporal fusion mechanism. Let $\mathbf{H} \in \mathbb{R}^{N \times T \times F_{out}}$ be the sequence of spatial embeddings generated by the GAT layers. The final prediction for node i fuses the most recent temporal state with the historical mean:

$$\hat{y}_i = \mathbf{W}_{fc} \left(\vec{h}_{i,T} + \lambda \frac{1}{T} \sum_{t=1}^{T} \vec{h}_{i,t} \right) + b_{fc} \quad (5)$$

where $\vec{h}_{i,T}$ is the embedding of the most recent month (acting as the strongest immediate predictor) and the average over time T acts as a stable historical residual to prevent sudden erratic spikes. The output is passed through an exponential inverse transformation to undo initial log-scaling, and then algorithmically converted into a normalized $0-100$ Risk Index for direct tactical deployment.

F. Geospatial Hotspot Detection (Secondary Engine)

While the RST-GAT handles proactive forecasting, the platform retains an optimized K-Means Clustering algorithm for rapid, retrospective dashboard visualization of historical high-density zones. The algorithm partitions the n observations into k sets $S = \{S_1, S_2, ..., S_k\}$ to minimize the within-cluster sum of squares (WCSS):

$$\arg\min_{\mathbf{S}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \quad (6)$$

where $\boldsymbol{\mu}_i$ is the geometric centroid of points in S_i. The model leverages the Elbow Method to dynamically determine the optimal k based on the specific filtered dataset loaded by the user.

G. Real-Time Surveillance Module

The active defense layer of the system utilizes a multi-threaded backend that toggles between two specialized computer vision architectures depending on the specific operational requirement.

1) Weapon Detection via YOLOv11: To identify immediate lethal threats, the system utilizes the YOLOv11 (You Only Look Once) architecture. YOLO frames object detection as a single regression problem, directly predicting bounding box coordinates and class probabilities simultaneously. The model optimizes a composite loss function involving bounding box regression (Complete Intersection over Union, or CIOU Loss) and focal loss for classification. This specific combination ensures high precision and recall on small, challenging objects like handguns or knives, even in low-light environments. The detector is deeply integrated with the ByteTrack algorithm, assigning and maintaining consistent unique IDs to threats as they move across consecutive frames.

2) Violence Recognition via MobileNetV2 + LSTM: Unlike static weapon detection which looks for specific objects, violence recognition must analyze human behavior over time. The engine uses a dual-stage temporal architecture. First, MobileNetV2 acts as a high-speed spatial feature extractor, stripping each video frame down to a 1280-dimensional feature vector, effectively ignoring background noise.

Sequences of 16 consecutive feature vectors are then fed into a Long Short-Term Memory (LSTM) network. The LSTM analyzes sequences of movement using internal gates to control information flow:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (7)$$
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (8)$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (9)$$
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (10)$$
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (11)$$
$$h_t = o_t * \tanh(C_t) \quad (12)$$

where f_t, i_t, and o_t represent the forget, input, and output gates respectively. This complex mathematical gating mechanism allows the network to successfully identify the erratic, high-kinetic sequences typical of physical violence while ignoring regular fast movements like running or dancing.

H. Public Perception Engine

To bridge the gap between objective crime statistics and subjective community sentiment, an NLP-driven engine was developed. The system queries the NewsAPI for local headlines based on user-selected geographical filters. The incoming text is parsed using the `TextBlob` library, which utilizes a built-in lexicon to evaluate the semantic polarity of thousands of words.

The algorithm averages the linguistic polarity of the recent articles, outputting a raw score ranging from -1.0 (Extremely Negative) to +1.0 (Extremely Positive). The backend mathematically maps this range into a normalized "Public Fear

Index" ranging from 0 to 100. Based on predetermined thresholds, the system visually tags the neighborhood's psychological status as Calm (0-40), Anxious (41-70), or Panic (71-100), providing law enforcement with crucial context regarding community relations.

I. Generative AI Decision Support

Google Gemini 2.5 Flash is integrated via an asynchronous API connection to act as the highest intelligence layer. This LLM powers a conversational Safety Assistant for real-time situational advice, evaluating the underlying tabular data to provide context-aware responses. Furthermore, an Automated Reporting Engine processes the predictive matrices and statistical summaries, generating highly professional, natural language executive reports formatted natively into PDF structures using `jsPDF` and `html2canvas`.

V. Implementation and Training Protocols

A. Hardware and Software Environment

The CrimeLens platform was developed, trained, and tested in a standardized, high-performance environment. Model training and performance validation occurred on systems equipped with an NVIDIA RTX 3060 GPU (6GB VRAM) running CUDA 12.1, paired with an Intel Core i5 processor and 16 GB of system RAM. The primary software stack utilized Python 3.11 for all backend services, with version control meticulously managed via Git.

B. Model Training and Hyperparameter Optimization

The PyTorch-based RST-GAT model underwent rigorous training cycles on municipal datasets spanning 5 years of recorded incidents. To prevent vanishing gradients during deep spatial aggregations, the model was constructed with specific hyperparameters:

- **Learning Rate:** Initiated at 0.001, controlled by a `ReduceLROnPlateau` scheduler with a patience of 10 epochs.
- **Attention Heads:** Restricted to 4 independent heads. Experimental trials with 8 heads exhibited rapid overfitting on localized, sparse crime regions.
- **Hidden Dimensions:** Node feature dimensions were capped at 64, providing a suitable bottleneck for regularization while maintaining expressiveness.
- **Dropout:** A rate of 0.2 was applied between the spatial convolution and temporal fusion layers to ensure generalized learning.
- **Loss Function:** Mean Squared Error (MSE) optimized via the AdamW algorithm.

C. Streaming and Asynchronous Workflows

A critical technical achievement of the platform is the thread-safe streaming of deep learning outputs to the React.js client. Video feeds generated by the computer vision surveillance module execute standard OpenCV `cap.read()` functions. The numpy arrays are rapidly passed through the YOLO/LSTM inference pipelines, annotated with bounding boxes, and yielded from the FastAPI server as encoded JPEG byte sequences using a `Multipart/x-mixed-replace` response boundary. This provides fluid, sub-30ms latency visual feedback directly in the browser without requiring external video hosting protocols.

VI. Results and Evaluation

The evaluation of the CrimeLens platform demonstrates a profound advancement in predictive precision through the implementation of the RST-GAT, alongside highly robust real-time security capabilities.

A. Spatio-Temporal Predictive Benchmarks

The flagship RST-GAT architecture was rigorously benchmarked against common industry alternatives on a massive hold-out test set of urban crime data. The models were evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) to gauge volumetric prediction accuracy, as well as a High-Risk F1-Score designed specifically to evaluate the model's ability to classify grids falling in the top 10% of danger zones.

TABLE I
Predictive Model Benchmark Comparison

Model Architecture	MAE	RMSE	High-Risk F1
RST-GAT (Proposed)	**0.3924**	0.5985	**0.6358**
XGBoost (Non-Spatial)	0.4375	0.5878	0.5122
Random Forest	0.4355	**0.5879**	0.4891
Linear Regression	0.4607	0.5950	0.3105
MLP (Ablation Baseline)	0.5428	0.7749	0.2450

As demonstrated in Table I, the graph-based RST-GAT model significantly outperformed the tabular baselines in pure spatial forecasting (MAE: 0.3924). Most importantly, the High-Risk F1 score of 0.6358 proves that the RST-GAT is vastly superior at identifying the absolute most dangerous grids—the exact metric critical for police resource allocation. While the XGBoost model achieved a slightly lower RMSE (suggesting fewer extreme outlier errors), its failure to capture topological risk flow rendered it much less effective for high-risk categorization.

An ablation study was also conducted, severing the graph adjacency matrix and forcing the model to rely solely on internal Multi-Layer Perceptrons (MLP). This resulted in a catastrophic drop in accuracy (MAE soaring to 0.5428), empirically validating that the spatial "ripple effect" captured by the graph attention mechanism is the most vital component of the system's predictive power. The RST-GAT successfully limits false positives by incorporating neighboring safety dynamics. When spatial autocorrelation is evaluated, the network effectively isolates micro-trends that standard models fail to distinguish.

B. Geospatial Visualization and UI Integrity

The platform successfully translates these complex multidimensional tensors into accessible, highly interactive tactical intelligence. The operational peak of the platform is the Spatial Risk Prediction Interface within the React dashboard. It dynamically renders the RST-GAT's normalized $0 - 100$ Risk Index over specific local city grids using Leaflet mapping tools, simultaneously outputting a sorted Priority Zone list to immediately direct patrol units.

For historical perspective, the interactive map utilizes continuous heatmap density layers integrated with the K-Means clusters (which achieved a strong Silhouette Score of 0.78 on operational datasets). To further expand visual comprehension, the `deck.gl` integration provides an immersive 3D Hexbin framework, aggregating spatial data into elevated hexagonal columns. This allows analysts to immediately identify vertical spikes representing major crime hotspots that otherwise appear flattened and unintelligible in traditional 2D overlays.

C. AI Surveillance Performance Metrics

The Real-Time Surveillance module achieved outstanding metrics during controlled testing environments designed to mimic public CCTV feeds.

TABLE II
REAL-TIME SURVEILLANCE ENGINE METRICS

Model / Task	Metric	Score	Latency (ms)
YOLOv11 (Weapons)	mAP@0.5	94.2%	∼25ms
YOLOv11 (Weapons)	Precision	97.1%	∼25ms
MobileNet+LSTM (Violence)	Accuracy	97.0%	∼40ms
MobileNet+LSTM (Violence)	AUC	81.71%	∼40ms

As detailed in Table II, the YOLOv11 weapon detection module achieved a Mean Average Precision (mAP) of 94.2%. By treating the detection process as a unified regression problem and leveraging ByteTrack logic, the model maintained 97.1% precision for small-scale weapons, successfully distinguishing handguns from visually similar objects like cellphones.

Simultaneously, the MobileNetV2 + LSTM violence engine achieved 97% accuracy on standard temporal benchmark datasets (e.g., Hockey Fight sequences). Against highly complex, crowded urban scenes (such as the UCF-Crime dataset), the hybrid temporal approach attained an AUC of 81.71%, generating reliable HUD overlays directly onto the video feed. Inference latencies across both models on dedicated GPU hardware hovered between 25 and 40 milliseconds, successfully sustaining a seamless 30 Frames Per Second (FPS) visual stream without frame dropping or client-side stuttering.

The Surveillance Monitor setup establishes an environment that acts as a dedicated application window where users select input sources and configure either the Weapon Detection or Violence Detection modes. The integration handles real-time streams seamlessly, rendering localized alerts directly via the frontend overlay.

D. Automated Intelligence Reporting

A critical administrative requirement for modern municipal deployment is the rapid translation of raw analytics into distributable, official documentation. Utilizing the integrated Gemini 2.5 Flash LLM, the system processes thousands of data points to synthesize a coherent, natural language executive summary. The platform seamlessly embeds this text alongside dynamic visual snapshots (such as Crime Severity breakdown charts generated by Chart.js) to compile a fully formatted PDF Crime Intelligence Report in under three seconds. This automation eliminates hours of manual administrative labor for police analysts. The resulting automated intelligence report includes a context-aware Executive Summary and embedded graphic snapshots providing a ready-to-distribute intelligence briefing.

E. Ethical Considerations and Bias Mitigation

Deploying predictive policing algorithms necessitates strict adherence to ethical standards. Predictive models have historically been criticized for creating feedback loops where over-policed neighborhoods generate disproportionate arrest data, subsequently driving the algorithm to continuously forecast risk in those exact areas. To mitigate demographic bias, the CrimeLens ingestion pipeline strictly operates on generalized incident reports and geographical matrices rather than individualized demographic variables. The explicit removal of race, socioeconomic status, and historical arrest rates from the primary tensor ensures the RST-GAT focuses strictly on the spatial flow of events rather than profiling individuals.

Furthermore, the introduction of the Public Perception Engine provides a necessary counterbalance. By constantly evaluating community fear levels through NLP sentiment analysis, the system prevents a pure "statistics-only" approach, giving law enforcement the tools needed to address the psychological wellbeing of the community alongside physical crime metrics.

F. Technical Limitations and Constraints

While the CrimeLens platform successfully meets its primary objectives for predictive spatial understanding and real-time threat engagement, specific technical limitations restrict its immediate global deployment:

- **Scalability Constraints (In-Memory Architecture):** The primary infrastructural constraint is the FastAPI backend's reliance on an in-memory Pandas DataFrame caching model to ensure rapid dashboard rendering. The system's capacity is strictly limited by the physical RAM of the hosting environment. Attempting to load datasets exceeding 1 GB into volatile memory triggers Out-of-Memory (OOM) errors, compromising the entire analytical engine.
- **Single-Session Persistence:** The platform currently lacks persistent data isolation and multi-user authentication databases, preventing its immediate use as a secure, multi-tenant production system for diverse police departments simultaneously.
- **Algorithmic Bottlenecks:** The flagship RST-GAT model requires complex adjacency matrix calculations. When generating spatial grids for massive urban datasets, the

real-time processing of these deep-learning tensors creates significant computational bottlenecks on non-GPU hardware.

VII. Conclusion and Future Scope

The CrimeLens project successfully demonstrates how deep graph-based technology and data-driven analysis can be utilized to understand and precisely forecast crime patterns. By transforming the urban landscape into an interconnected network of topological nodes, the flagship RST-GAT architecture achieved a breakthrough MAE of 0.3924, significantly outperforming traditional tabular baseline models that fail to capture spatial dependencies. Complementing this macro-level forecasting, the integration of high-speed YOLOv11 and MobileNetV2+LSTM computer vision models transitioned the platform from a purely historical reporting tool into an active, real-time defense layer capable of immediate threat recognition. Finally, the inclusion of Generative AI and NLP sentiment algorithms bridged the gap between raw statistical data and the reality of human community safety.

The future scope of CrimeLens is centered on evolving the platform from a sophisticated proof-of-concept into a highly scalable, live intelligence ecosystem capable of continuous, multi-tenant collaboration. A primary engineering objective is replacing the current in-memory data storage with a persistent, spatial database solution such as PostgreSQL equipped with PostGIS. This transition will eliminate volatile memory limits, enabling the continuous ingestion of live municipal data directly into the RST-GAT framework.

Furthermore, future iterations of the platform will dramatically expand the graph neural network. The objective is to incorporate multi-modal external data nodes—such as live weather feeds, public transit schedules, and major city event calendars—directly into the adjacency matrices. This will allow the system to capture even deeper socio-temporal dependencies, further boosting forecast accuracy and providing law enforcement with the ultimate tactical agility required for modern urban safety management.

References

[1] E. Cesario, P. Lindia, and A. Vinci, “Multi-density crime predictor: an approach to forecast criminal activities in multi-density crime hotspots,” *J Big Data*, vol. 11, no. 75, 2024.

[2] E. G. Ilgun and M. Dener, “Exploratory data analysis, time series analysis, crime type prediction, and trend forecasting in crime data using machine learning, deep learning, and statistical methods,” *Neural Comput & Applic*, vol. 37, pp. 11773–11798, 2025.

[3] N. Shah, N. Bhagat, and M. Shah, “Crime forecasting: a machine learning and computer vision approach to crime prediction and prevention,” *Vis. Comput. Ind. Biomed. Art*, vol. 4, no. 9, 2021.

[4] J. Wu, S. M. Abrar, N. Awasthi, et al., “Enhancing short-term crime prediction with human mobility flows and deep learning architectures,” *EPJ Data Sci.*, vol. 11, no. 53, 2022.

[5] J. Bonam, L. R. Burra, G. S. V. N. S. Susheel, K. Narendra, M. Sandeep, and G. Nagamani, “Crime Hotspot Detection using Optimized K-means Clustering and Machine Learning Techniques,” *2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC)*, Coimbatore, India, 2023, pp. 787–792.

[6] B. G and K. G, “Crime Hotspot Classification Using Machine Learning,” *2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS)*, Prawet, Thailand, 2025, pp. 220–228.

[7] M. F. Buland Iqbal et al., “A Comparative Analysis of Multiple Methods for Predicting a Specific Type of Crime in the City of Chicago,” *arXiv preprint arXiv:2304.13464*, 2023.

[8] A. Khan, N. Chandran, and D. Gangodkar, “Geospatial analysis of crime patterns and hotspots for crime prevention using deep learning algorithms,” *EDPACS*, pp. 1–14, 2025.

[9] V. Mandalapu, L. Elluri, P. Vyas, and N. Roy, “Crime prediction using machine learning and deep learning: A systematic review and future directions,” *arXiv preprint arXiv:2303.16310*, 2023.

[10] P. Maurya, T. Shaikh, I. Ahmed, A. Firdosi, and K. Deshmukh, “Crime analysis and prediction using machine learning,” *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 2023.

[11] S. Vosta and K. C. Yow, “A CNN-RNN Combined Structure for Real-World Violence Detection in Surveillance Cameras,” *Applied Sciences*, vol. 12, no. 3, p. 1021, 2022.

[12] A. Thakur, A. Shrivastav, R. Sharma, T. Kumar, and K. Puri, “Real-Time Weapon Detection Using YOLOv8 for Enhanced Safety,” *arXiv preprint arXiv:2410.19862v1*, 2024.

[13] H. M. B. Jahlan and L. A. Elrefaei, “Detecting Violence in Video Based on Deep Features Fusion Technique,” *Computer Science Department, King Abdulaziz University*, 2019.

[14] S. Vosta and K. C. Yow, “A CNN-RNN Combined Structure for Real-World Violence Detection in Surveillance Cameras,” *Applied Sciences*, vol. 12, no. 3, 1021, 2022.

[15] G. Jin et al., “Spatio-Temporal Graph Neural Networks for Predictive Learning in Urban Computing: A Survey,” in *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 8, pp. 3828-3847, 2024.

[16] X. Wang, L. Zhang, and H. Chen, “Spatio-Temporal Graph Neural Networks for Accurate Crime Prediction,” *2023 IEEE International Conference on Data Mining (ICDM)*, 2023, pp. 1205-1210.